Explainable AI Clinical Decision Support System

Clinical Prediction Report

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Patient Information

Patient ID	PDF002
Patient Name	Heart Report Test
Doctor	Test Doctor
Hospital	Test Hospital
Disease	Heart Disease
Age	63
Gender	Male
Chest Pain Type	Non-anginal Pain
Resting Blood Pressure	145 mmHg
Cholesterol	233 mg/dL
Fasting Blood Sugar	Yes
Rest ECG	Normal
Maximum Heart Rate	150
Exercise Angina	No
ST Depression	2.3
ST Segment Slope	Upsloping
Major Vessels	0
Thalassemia	Fixed Defect

Prediction Result

Prediction	No Heart Disease
Probability	31.0 %

Risk Level	Low

Clinical Recommendation

No significant indication of heart disease was detected. Maintain a healthy lifestyle, exercise regularly, and undergo routine medical check-ups.

Top Influencing Features (SHAP)

Feature	Impact
Chest Pain Type	-0.098
Major Vessels	-0.0853
Thalassemia	0.074
Slope of ST Segment	-0.0553
ST Depression	0.0418

LIME Explanation

Feature	Impact
cp <= -0.16	-0.156
oldpeak > 0.54	0.0503
-0.88 < thal <= 1.18	0.0491
exang <= -0.70	-0.033
-0.74 < chol <= -0.17	0.0243

Generated by Explainable AI Clinical Decision Support System

Machine Learning • SHAP • LIME • Flask • React

Doctor Signature: _____

Patient Signature: _____